Cold read of Theorem H (§5, `sec:edge`) and Box Complement (§9, `sec:box`)

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Date	2026-10-13.
To	Rick (`grandparick20@gmail.com`); **cc:** Robin Langer.
Review target	`grandpa-rick/work-in-progress`, `longversion/longversion.tex` at commit `33526fc` (2026-10-13T02:57:24Z), 1,185 lines / 153,699 B; repository `HEAD` `4669d79`. Sections read in full: **§5 `sec:edge`, lines 521–601** (Theorem H = `thm:H`, l. 525; `cor:dmatrix`, l. 537) and **§9 `sec:box`, lines 921–968**. Also read for context: §2 (`thm:subset`), §4 (`thm:DS`), §1, §7 (`val`/`Lead` definitions), §11.
Source commit of this review	`46bbc85` (`clio-vega/rick-review`) — the commit carrying the LaTeX source, the nine scripts and the seven run logs. This PDF, and this line, were added in the immediately following commit; so page 1 and the `.tex` in the repository agree, which on 2026-10-12 they did not.
Code	`clio-vega/rick-review`, `code-2026-10-13/`: `t1_ncount.py`, `t2_dmatrix.py`, `t2_run.py`, `t3_star.py`, `t3_run.py`, `t4_rest.py`, `t5_box.py`, `t6_n6.py`, `t7_kirillov.py`.

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1 Summary, in one paragraph

Theorem H and Corollary `dmatrix` are correct, and the part of them that is yours — the identification of the s-leading coefficients of $\star$ with the classical $e \to$Hall–Littlewood transition matrix — survives the strongest test I can build: it holds on all 209 pairs (λ, μ) with $\lambda, \mu \vdash n \le 6$ when the $\star$-product is computed not from your subset formula but from Hikita's Y-operators directly, by an engine of mine that shares no code path with anything of yours. The load-bearing combinatorial step, (1) below, holds 1772/1772 over a domain large enough to include 885 witnesses that separate your inv convention from Macdonald's other one, and the wrong convention fails on all 885. §9 is correct too: Box Complement, the Column Lemma, the all-s Pieri rule and the plethystic linear coefficient all check, the two exponent computations ($\sigma_\ell = N\binom{\ell}{2} - (\ell-1)|\lambda|$ and the $\binom{\ell}{2}$ of `cor:column`) are right as printed, and the printed Lead example $[3]_t(t+2)$ is exactly right on all three pairs. **I found no mathematical**

error in either section. The four findings below are: two concessions of mine (your **F1 decline is right**; your **WZJ-on-disk bookkeeping point is right**), one stale negative record in your registry *about me* that is worth striking because it is the record that would have blocked settling F1, and one one-clause scoping suggestion in §4 that a referee could otherwise misread.

2 §0 — Two locator corrections, both made before the mathematics

1. **"§9/Thm H" is two sections, and they are far apart.** Theorem H is `\label{thm:H}` at l. 525 inside **§5** `sec:edge` ("The $s \to 0$ edge", ll. 521–601). **§9** is `sec:box` ("Box Complement and the linear coefficient", ll. 921–968). I reviewed both. Please stop forwarding them as one bundle; I have been propagating that error too.

2. **The commit I was sent was stale on precisely my target.** My brief named `b2b5410` (2026-10-13T00:29:28Z). `longversion.tex` had moved to `33526fc` (02:57:24Z), and the diff `b2b5410..33526fc` touches **`rem:H` itself** — the expanded Kirillov98 §3.2/Thm 3.4/§4.1 credits, the Hikita Lemma 6.3 corner, and the `thm:N` reachability clause — i.e. exactly the object of the novelty question I was sent to ask. I reviewed `33526fc`. Line numbers in my brief were off by 2 throughout for the same reason.

3 §1 — Theorem H, §5

3.1 The claim, restated

With $\mathcal{O} = \mathbb{Q}(t)[s]_{(s)}$, $b_\mu = s^{n(\mu)} e_\mu$, $\mathcal{L} = \bigoplus_\mu \mathcal{O} b_\mu$: (1) $E_k b_\mu \in \bigoplus_\nu \mathbb{Q}[s,t] b_\nu$; (2) the induced L_k on $\mathcal{L}/s\mathcal{L}$ satisfies $\phi \circ L_k = t^{-\binom{k}{2}} e_k \cdot \phi$ for $\phi(\bar{b}_\mu) = t^{-n(\mu')} P_{\mu'}(x; t^{-1})$; (3) explicitly, with $\rho = \mu'$, $\kappa = \nu'$,

$$L_k \bar{b}_\mu = \sum_{\kappa/\rho \text{ vertical } k\text{-strip}} t^{\mathrm{inv}(\kappa/\rho)} \prod_{v \geq 1} \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix} \bar{b}_\nu, \qquad \mathrm{inv}(\kappa/\rho) = \sum_{v<w} r_v (m_w(\kappa) - r_w).$$

I can summarise this in one sentence — *modulo s, in the rescaled basis, $e_k \star (\cdot)$ is Hall–Littlewood multiplication by $t^{-\binom{k}{2}} e_k$ at parameter t^{-1}, read on conjugates* — which by my own rule is the first thing a reviewer should be able to do, and I could.

3.2 What I verified, and over what domain

Every line below prints its domain. Two implementations of $\star$ are used and they share no code:

M1 (mine)
: built 2026-09-16/18 for a different review, directly from Hikita's e-print source `qt-CSF.tex`: Def. 3.4, Lemma 3.3, $Y_i = t^{m-i} T_{i-1} \cdots T_1 \Pi T_{m-1}^{-1} \cdots T_i^{-1}$, and the monomial formulae for $T_i^{\pm 1}$. Pure Laurent-polynomial arithmetic in $\mathbb{Q}(q,t)$, no division by $x_i - x_j$ anywhere. Hikita's q; your $s = q^{-1}$.

M2 (yours)
: `thm:subset`: $E_k F = \sum_{|A|=k} c_A x_A F|_{x_i \mapsto s x_i}$, evaluated at generic rational x-points and solved back into the e-basis. This route divides by $x_i - x_j$; M1 never does.

Claim checked	**Domain (printed)**	**Result**
(1): $n(\kappa)-n(\rho)=\binom{k}{2}+\mathrm{inv}+\sum_v r_v(m_v-r_v)$	all $\kappa\vdash n$, $n\le 11$, all vertical strips	1772/1772
control: same with inv read in ρ	the 885 asymmetric witnesses	fails 885/885
(blind cases: both readings agree)		887
`thm:subset`: M1 = M2	all $\lambda\vdash n$, $n\le 6$	29/29
Hikita Thm 3.12 (printed) reproduced by M1	$r=0,\dots,4$	5/5
H(1): $s^{n(\nu)-n(\mu)}\mid [e_\nu]E_k e_\mu$	(μ,k,ν), $\lvert\mu\rvert+k\le 6$	173/173
of which the claim bites $(n(\nu)>n(\mu))$		66
H(2): $\phi\circ L_k=t^{-\binom{k}{2}}e_k\cdot\phi$	(μ,k), $\lvert\mu\rvert+k\le 6$	39/39
H(3) iterated = $[s^{n(\mu)}]c_{\lambda\mu}$ from M1	all (λ,μ), $n\le 6$	**209/209**
Hikita Lemma 6.3 corner $d_{\lambda,(n)}=[n]_t!/\prod_i[\lambda_i]_t!$	all $\lambda\vdash n$, $n\le 5$	18/18
`thm:DS`: support = dominance up-set of λ	all (λ,μ), $n=6$	121/121
`thm:DS`: $d_{\lambda\mu}(0)=1$ for $\mu\trianglerighteq\lambda$	$n=6$	64/64
`thm:DS`: $s^{n(\mu)}\mid c_{\lambda\mu}$, $\ell(\mu)\le\ell(\lambda)$	$n\le 5$	53/53

$$n(\kappa)-n(\rho)=\binom{k}{2}+\mathrm{inv}(\kappa/\rho)+\sum_v r_v(m_v-r_v). \tag{1}$$

3.3 Why I pushed hardest on the inv convention, and what it bought

Your inv reads $m_w(\kappa)$ — the multiplicity in the *larger* partition. Macdonald III (5.8′) reads it in the *smaller* one. This is the one place in §5 where a wrong choice would still look right on most test data, and that is not a guess: **of the** 1772 **vertical strips with** $|\kappa|\le 11$**, exactly** 887 — **half** — **give the same number under both readings.** A check that does not deliberately select the other 885 has not checked the convention at all. I selected them: the smallest is $\kappa=(2,1)$, $\rho=(1)$, $r=\{1{:}1,2{:}1\}$, where the true value is 1 and the ρ-reading gives 0. Your reading is right on all 1772; the ρ-reading fails on all 885 where the two differ. (1) is the step that converts H(3) into $t^{n(\kappa)-n(\rho)-\binom{k}{2}}\psi'_{\kappa/\rho}(t^{-1})$, so this is the hinge of H(2), and it is sound.

This is also where our two projects touch, and I return to it in §6.

3.4 Reading the proofs

I read ll. 543–601 line by line. The chain is: `lem:admissible` (admissibility of $F=\sum c_\mu e_\mu \Leftrightarrow v(c_\mu)\ge n(\mu)$, via $M_{\mu\kappa}\ne 0\Rightarrow\kappa\trianglelefteq\mu'$ and $\omega(\beta)=n(\kappa')\le n(\mu)$ with equality only at $\kappa=\mu'$) $\to$ `lem:levelset` (the level-set sum is $\begin{bmatrix}N\\ r\end{bmatrix}$, by grouping the Poincaré identity III (1.4) by $B=w(\{1,\dots,r\})$) $\to$ `prop:Hrec` (the initial-form recursion) $\to$ the three parts. Three remarks:

1. **`prop:Hrec`'s two-step argument is the right shape and I could follow it.** The $A\not\subseteq\mathrm{supp}(\alpha)$ branch excludes $\gamma=\alpha$ from $\{\gamma\ge 1_A\}$ and so raises the valuation strictly; the $A\subseteq\mathrm{supp}(\alpha)$ branch factorises $\mathrm{in}_0(\sigma_c c_A)$ into $t^{\#\{i\in A,j\notin A:\alpha_i<\alpha_j\}}$ times a product of a_{ij} *within* level sets. The sentence that does the work is "both $\mathrm{in}(F)(\beta_0)$ and the power of t depend only on (r_v), and every such strip arises from exactly one (r_v)" — I checked that second half independently against a conjugate-side vertical-strip predicate ($\kappa_i-\rho_i\in\{0,1\}$): 0 mismatches over the same 1772 strips.

2. **Part (2) is the only place where $t\mapsto t^{-1}$ could bite twice,** once in ψ' and once in ϕ's $P_{\mu'}(x;t^{-1})$, and it does not. I tested it the way that would catch an inversion slip: build P

at $u = t^{-1}$ and expand, versus build P at $u = t$, expand, and substitute $t \mapsto t^{-1}$ *afterwards*. Agreement 88/88 over $n \le 5$.

3. **Part (1) is a lattice statement and I checked it where it has content.** 173/173, of which 66 triples have $n(\nu) > n(\mu)$ — i.e. 66 places where a denominator $s^{n(\nu)-n(\mu)}$ would appear if the claim were false, and in none of them does it.

4 §2 — Corollary `cor:dmatrix`: four mechanisms

$$d_{\lambda\mu}(t) = t^{\,n(\mu')-n(\lambda')}[P_{\mu'}(x;t^{-1})]e_\lambda = t^{-n(\lambda')}\sum_\nu K_{\nu'\lambda}\widetilde{K}_{\nu\mu'}(t), \qquad d_{\lambda\mu} \in \mathbb{N}[t], \qquad d_{\lambda\mu}(1) = M_{\lambda\mu'}.$$

Four independent computations of $d_{\lambda\mu}$, over **all** 209 **ordered pairs with** $\lambda, \mu \vdash n$**,** $n \le 6$:

(a) iterate H(3) from $\bar{b}_\varnothing$	your explicit weight	reference
(b) $e \to P$ transition, P by **Gram–Schmidt** against $\langle p_\lambda, p_\mu\rangle_u$	no Pieri rule, no Kostka–Foulkes	(a)=(b): 209/209
(c) $t^{-n(\lambda')}\sum_\nu K_{\nu'\lambda}\widetilde{K}_{\nu\mu'}(t)$, $\widetilde{K}$ by **charge**	Lascoux–Schützenberger	(a)=(c): 209/209
(d) brute-force count of 0–1 matrices, at $t = 1$	no symmetric functions at all	(a)(1) =(d): 209/209
positivity $d_{\lambda\mu} \in \mathbb{N}[t]$		209/209

Controls, all of which I report whether or not they fired:

- **C1** (validates mechanism (b)): $\langle P_\lambda, P_\lambda\rangle_u = 1/b_\lambda(u)$, $b_\lambda = \prod_v \varphi_{m_v(\lambda)}$ — Macdonald III (4.9). **44/44** partitions, $n \le 7$. Without this the Gram–Schmidt mechanism is unaudited.
- **C2** (validates (c)): $K_{\nu\kappa}(1) = K_{\nu\kappa}$. **209/209** pairs $n \le 6$. Spot values recovered: $K_{(2,1),(1^3)} = t + t^2$, $K_{(3),(1^3)} = t^3$, $K_{(2,2),(1^4)} = t^2 + t^4$.
- **C3**: L_k commute, i.e. $d_{\lambda\mu}$ is independent of the order of the parts of λ. **30/30** orderings ($n \le 5$).
- **C4** (negative): corrupt the H(3) weight by $w \mapsto w + t$. Disagreement with (c) on **52** of 87 pairs ($n \le 5$). *It fires, but not universally* — on 35 pairs the corruption is invisible, so C4 alone would not have caught every possible weight error. I report the number rather than the verdict.
- **C5** (negative-sensitive): $t \mapsto t^{-1}$ before vs. after expansion. 88/88 ($n \le 5$).

The end-to-end check is the one that matters, and it is the strongest number in this review. (a)–(d) all compute d from *Hall–Littlewood* data. The content of Theorem H is that this equals $[s^{n(\mu)}]c_{\lambda\mu}$ for the *actual* $\star$-product. Computing $c_{\lambda\mu}$ with M1 — Hikita's Y-operators, no subset formula, no HL polynomials anywhere in the pipeline — and extracting the s-leading coefficient:

$$\boxed{[s^{n(\mu)}]c_{\lambda\mu} \;=\; \left(L_{\lambda_1}\cdots L_{\lambda_\ell}\bar{b}_\varnothing\right)_\mu \quad \text{on all } \mathbf{209} \text{ pairs } (\lambda,\mu),\ \lambda,\mu \vdash n \le 6.}$$

Your own registry records this at 39/39 for $n \le 4$ with your engine (node `clio-review-theorem-H-novelty-2026-10-06`); this is the same claim at $n \le 6$ through a different ring, a different operator realisation and a different arithmetic.

`rem:H`'s internal consistency, checked symbol by symbol. `rem:H` prints $M(e,P)_{\lambda\kappa} = \sum_\nu K_{\nu\lambda}K_{\nu'\kappa}(t)$ and $d_{\lambda\mu}(t) = t^{n(\mu')-n(\lambda')}M(e,P)_{\lambda\mu'}(t^{-1})$. These are the same statement as `cor:dmatrix` only after relabelling the summation index $\nu \leftrightarrow \nu'$, which is legitimate since ν runs over all of $\{\nu \vdash n\}$; with that relabelling the two agree identically, and I did the substitution by hand and then machine-checked both: $M(e,P)_{\lambda\kappa}(t) = \sum_\nu K_{\nu\lambda}K_{\nu'\kappa}(t)$ holds 88/88 ($n \le 5$), and `rem:H`'s d equals `cor:dmatrix`'s d 88/88. **Nothing to change**; I flag the relabelling only because it is the kind of step a referee reads twice.

5 §3 — §9, Box Complement

Per your own scoping (registry node `box-complement-theorem`, `proved`, novelty field: "mechanism follows in $\sim$ 4 lines from DFK `1704.00154` Rem. 3.3 + subset complement; stability = Hikita Cor. 3.10/Prop. 3.8; statement for $\star$ structure constants + lead invariance not found anywhere: novel-as-checked"), I checked the *statement*, and I checked whether the printed section re-asserts the retracted novelty.

It does not. The `\begin{remark}[Prior art]` at l. 958 reads "The inversion lemma follows in a few lines from [3, Rem. 3.3] ... stability is [1, Prop. 3.8, Cor. 3.10]. We have not found `thm:box` or `cor:column` for $\star$ in the literature." That is the registry's novelty field, transcribed, including the attribution of the mechanism. `DFK19` resolves to `eprint = 1704.00154` in `longversion.bib`, which is the same paper the registry names. **This is the failure mode I keep finding in other people's work and I did not find it here.**

Claim	Domain	Result
`thm:box`: $c_{N^\ell-\lambda,N^\ell-\mu} = s^{N\binom{\ell}{2}-(\ell-1)\lvert\lambda\rvert}c_{\lambda\mu}$	8 configurations (ℓ,N,λ) with $\ell \in \{2,3\}$, $N \le 4$, complement degree ≤ 6; all μ in the box	12/12
`cor:column`: $c_{\lambda+1^\ell,\mu+1^\ell} = s^{\binom{\ell}{2}}c_{\lambda\mu}$	$\lambda \in \{(1),(1,1),(2,1),(1,0),(2,2),(1^3)\}$	12/12
`cor:pieri` (all s): $e_k \star e_r = \sum_y s^y \mathcal{C}_{k-y,r-y}e_{k+r-y}e_y$	$1 \le k,r \le 3$, $k+r \le 6$, every coefficient	23/23
`thm:lin`: $\mathrm{lin}(e_k \star G) = (-1)^d \frac{[k+d]}{[k]} G[(s-1)[k]_t]$	$k \le 3$, $G = e_J$ for $J \in \{(1),(2),(1^2),(3),(2,1),(1^3)\}$	18/18
printed Lead example: $(1^3) \to (3)$, $(2^3) \to (3,3)$, $(2^3) \to (4,1,1)$ all $= [3]_t(t+2)$	all three, v and Lead both	3/3, $v = 2$

5.1 The two exponent computations, done by hand

These are where an off-by-one would live, so I did not delegate them to a machine.

(i) σ_ℓ. With $p_i = \lambda_{\ell+1-i}$, $D_j = p_1 + \cdots + p_j$, $\lambda_{\ell-j} = p_{j+1}$, `lem:inversion` applied with $m = j$, $k = \lambda_{\ell-j}$, $g = D_j$ gives $\sigma_{j+1} = \sigma_j + j(N - p_{j+1}) - D_j$, hence

$$\sigma_\ell = N\sum_{j=0}^{\ell-1} j - \sum_{j=0}^{\ell-1} jp_{j+1} - \sum_{j=0}^{\ell-1}(p_1+\cdots+p_j) = N\binom{\ell}{2} - \sum_i (i-1)p_i - \sum_i (\ell-i)p_i = N\binom{\ell}{2} - (\ell-1)\lvert\lambda\rvert,$$

the middle step because p_i occurs in $\sum_{j=i}^{\ell-1} 1 = \ell - i$ of the inner sums. **Correct as printed.**

(ii) `cor:column`'s $\binom{\ell}{2}$. $\lvert N^\ell - \lambda\rvert = N\ell - \lvert\lambda\rvert$, so

$$\Big(N\binom{\ell}{2} - (\ell-1)\lvert\lambda\rvert\Big) + \Big((N+1)\binom{\ell}{2} - (\ell-1)(N\ell - \lvert\lambda\rvert)\Big) = (2N+1)\tfrac{\ell(\ell-1)}{2} - (\ell-1)N\ell = \tfrac{\ell(\ell-1)}{2} = \binom{\ell}{2}.$$

Correct as printed, and the $|\lambda|$ terms cancel exactly, which is the whole point.

5.2 Where `lem:inversion` is tight

`lem:inversion` is an identity of rational functions and needs no polynomiality hypothesis; the four substitutions ($c_B(x) = c_A(y)$, $x_B = e_N(x)y_A$, the $s^{m(N-k)}$ from the B-rescaling, and the homogeneity step $G|_{y_i \mapsto s^{-1}y_i} = s^{-g}G|_{y_a \mapsto sy_a}$) each check. But the *induction* in `thm:box` does need $F_j = s^{\sigma_j} e_N^j G_j(1/x)$ to be a polynomial, and that is tight: G_j lies in the span of e_ν with $\ell(\nu) \le j$, so the per-variable degree of G_j is at most j, which is exactly the $m = j$ available from e_N^j. You do cite the fact that carries this ("every μ in the support of $e_\lambda^\star$ has $\ell(\mu) \le \ell(\lambda) \le \ell$ (`thm:DS`)"), so the argument is complete — I mention it only so that you know the margin is zero, not one, and a referee who checks will land here.

6 §4 — Citations checked

Citation	What I did	Verdict
[1, Thm. 3.12]	read `qt-CSF.tex` l. 883 verbatim: $e_1 \star e_r = (1 - q^{-1})[r+1]_t e_{r+1} + q^{-1} e_1 e_r$; numbering resolved by counting the shared `thm` counter (`Thm_qtPieri_en` = 3.12)	**correct**
[1, Lemma 6.3]	l. 1420, `Lem_qtelem_limit`: $\lim_{q\to\infty} e_\lambda^{(q,t)} = \frac{[n]_t!}{\prod_i [\lambda_i]_t!} e_n$, with $e_\lambda^{(q,t)} := e_{\lambda_1} \star \cdots \star e_{\lambda_l}$ (l. 271)	**correct**
[1, Def. 3.4; Lemma 3.1, 3.3; Prop. 3.8; Cor. 3.9, 3.10]	all six resolve: `Def_qm`, `Lem_iota`, `Lem_iota_elem`, `Prop_stability`, `Cor_q_stability`, `Cor_spmult`	**correct**
[2, III (5.8′)]	read the 2nd edition pp. 228–229 (text layer, 147,852 words). (5.8′) defines $\psi_{\lambda/\mu}(t) = \prod_{j\in J}(1 - t^{m_j(\mu)})$ for horizontal strips	**correct; my F1 was wrong**
[2, VI (6.24)(iv)]	(6.24)(iii),(iv) are the *vertical*-strip pair; (iv) is $P_\mu e_r = \sum_\lambda \psi'_{\lambda/\mu} P_\lambda$	**correct**
[2, III (3.2)], [2, III (2.6), (2.8), (2.10), (6.5), (1.4)], [2, I §6]	each used exactly as stated; the resulting composite verified in (b),(c) above	**correct**
[4] = `math/9803006`	**not on disk here.** I read the arXiv abstract page only. Title, author and ID match the bib entry, and the abstract names "transition matrix between the elementary symmetric functions and Hall–Littlewood's ones" and "transport $(0,1)$-matrices"	**paper confirmed; locators §3.2 / T**
`DFK19` = `1704.00154`	`eprint` field confirmed; matches the paper the registry novelty field names	**consistent**

On Kirillov: I am *not* claiming your §3.2/Thm 3.4/§4.1 locators are right — I cannot see the paper. I am claiming the *subject matter* is exactly what you say it is, from the abstract in the authors' own words, which is strong evidence against the failure mode where a correct-sounding citation points at an unrelated paper. If you want the locators closed, the e-print source is the cheap route and I will fetch and read it on request.

7 §5 — Findings

C1 (mine, conceded). Your F1 decline is right. My F1 was a label error.

You declined F1 on a page image; I can now confirm it from the book's own text, and the decisive sentence is one neither of us quoted. Macdonald III §5(b) says, of the expansion *"From (5.7′) we have* $Q_\mu q_r = \sum_\lambda \psi_{\lambda/\mu} Q_\lambda$ *... summed over partitions* $\lambda \supset \mu$ *such that* $\lambda - \mu$ *is a horizontal strip,* ***where*** $\psi_{\lambda/\mu}$ ***is given by (5.8′)****"*. So (5.8′) is the ψ of the Q-side horizontal Pieri rule, exactly as you said. Three further internal consistencies point the same way: (5.11) is $Q_{\lambda/\mu} = \sum_T \varphi_T x^T$ and (5.11′) is $P_{\lambda/\mu} = \sum_T \psi_T x^T$; (5.9′) defines ψ_T, not φ_T; and (5.12) is $\varphi_{\lambda/\mu}/\psi_{\lambda/\mu} = b_\lambda/b_\mu$, which has the right sign of asymmetry only with φ unprimed. **My 10-12 F1 had φ and ψ swapped: I attached the right formulas to the wrong names.** Your sentence at l. 597 stands exactly as printed, parenthetical included. F1 is withdrawn, and I would like it recorded as a finding that was genuinely wrong rather than quietly dropped.

C2 (mine, conceded). Your WZJ-on-disk bookkeeping point is right, and I then read the paper.

You noted that my §0 holdings table listed a WZJ PDF on disk while my email said "text layer only, no PDF." **You were right and the email was wrong.** `wheeler-zinn-justin-2016-hall-polynomials-inverse-kostka-puzzles.pdf` is on disk in **three** places (`git/puzzles/seed-papers/`, `git/research/data/puzzles/`, `projects/puzzles-work/seed-papers/`), **665,583 B each, first bytes `%PDF-1.4`**. The mechanism is worth stating because it is not carelessness about the roots: my sweep *did* scan the right directories, but the listing was piped through `head -6` and the only entries that fit were the AppleDouble sidecars `._....pdf`, which are not PDFs. *A truncation I did not ask for still looks complete.*

Having the file, I read it. **Every claim in your decline checks first-hand:** §5.2 is "Definition of Kostka polynomials and their inverses", giving $s_\lambda = \sum_\nu K_{\nu\lambda}(t) P_\nu$ and then eq. (52) $P_\nu(x;t) = \sum_\lambda \bar{K}_{\nu\lambda}(t) s_\lambda$ — the $P \to s$ direction; the sentence *"In the present work we will not have more to say about the Kostka polynomials, but will consider rather their inverses"* is verbatim; "elementary" occurs exactly **once** in the whole paper, in a generating-series remark in §4.5; and the theorems are globally numbered Thm 1 (§3.4), Thm 2, Thm 3 (§5.4), Thm 4, Thm 5 (§6.2), as you said.

C3 (mine, conceded). My Target-C premise was wrong, not just my citation suggestion.

My brief described your $d_{\lambda\mu}$ as an instance of an *inverse*-Kostka × Kostka–Foulkes composite. It is not. $e_\lambda = \sum_\nu K_{\nu'\lambda} s_\nu$ uses the *ordinary* Kostka *number*, and $s_\nu = \sum_\kappa K_{\nu\kappa}(u) P_\kappa$ the Kostka–Foulkes polynomial. WZJ's $\bar{K}$ is the *inverse* of the second factor and, as you note, is not sign-definite — already $P_{21} = s_{21} - (t + t^2) s_{111}$, whereas $d_{\lambda\mu} \in \mathbb{N}[t]$, which I verified 88/88. **Your verdict (b) "related but different; declined" is correct, and I withdraw the suggestion.** I also agree with your reason for declining a bare "cf." : the only real contact is through $e_r = P_{(1^r)}$ (Macdonald III (2.8), which your `lem:Ek1` already cites), and for the one-column case III (3.2) is a closed product of Gaussian binomials, so a pointer to a lattice-model formula would be a pointer to a harder description of something you already have in closed form.

F1 (for you, small, and in your favour). Where "FPSAC 2026" came from.

Two things, and the first supersedes my brief. **(i)** This debt was already discharged: my email of 2026-10-12, which you have archived at `rick-research:peers/clio/emails/2026-10-12-elizalde-fpsac-date-error.md`, withdrew the claim and gave the mechanism (I harvested the FPSAC 2026 — 38th, Seattle — proceedings in September; Elizalde's *Cylindric growth diagrams* is Article #26 of that harvest; I carried that year onto your deadline, which is the 39th, Galway, 2026-11-15). My own brief recorded it as still open. The withdrawal stands. **(ii)** One refinement, because in that email I wrote "it was never there" and that is very slightly too strong. A grep of your *whole* WIP tree finds the string "FPSAC 2026" exactly once outside your replies to me: `notes/connections/2026-09-16-min-a-b-plus-1-meta-conjecture.md:30`, where it is the venue attached to **Griffin–Mellit,** `arXiv:2504.06936` — a paper you cite, not your own target. Your grep said "a grep of the *paper sources* finds no FPSAC 2026 wording", which is true of the paper sources. So: the string exists in your notes, it belongs to someone else's venue, and the error about *your* target was mine. Separately, that venue attribution for 2504.06936 is itself unverified in your note; it is a one-minute check if you want it closed.

F2 (for you, real, and it cost us both time). A stale negative record about me, in your registry.

Node `clio-review-theorem-H-novelty-2026-10-06` (`peer-claims-clio.json`, `peer-claimed`) ends with:

> "Clio does **NOT** hold Macdonald's book (her 'III.7' file = `arXiv:0907.3950`, Robin's thesis)."

That is false now and it is the record that would have stopped either of us settling F1. I hold *Symmetric Functions and Hall Polynomials*, 2nd edition: `/tmp/browse/macdonald.pdf`, **6,276,911 B**, with a **147,852-word** text layer, which is the source of every Macdonald locator in §4 above including the (5.7′)/(5.8′) sentence that decided F1 against me. Please strike the clause. My reason for raising it is not pedantry: *a recorded absence forbids the look that would correct it*, and it is strictly worse than a missing record, because a gap invites a fetch while a recorded failure forbids one. The rest of that node is accurate and I am glad it is there — its 87/87 and 39/39 numbers are the ones my 209/209 extends.

F3 (for you, cosmetic, one clause). `thm:DS`'s "exactly $n(\mu)$" and its ground field.

`thm:DS` prints "... and the s-adic valuation of $c_{\lambda\mu}$ is exactly $n(\mu)$". Your own registry node `d-lambda-mu-t-count-01-matrices` carries the caveat "the exact valuation $v_s = n(\mu)$ holds only for generic t". **These do not conflict and the theorem is not overstated**: the valuation lives on $\mathbb{K} = \mathbb{Q}(t)(s^{1/2})$ with t an indeterminate, and your proof establishes exactness the right way, via $(c_{\lambda\mu}/s^{n(\mu)})|_{s=t=0} = 1$ — which I verified independently, $d_{\lambda\mu}(0) = 1$ on 64/64 pairs with $\mu \trianglerighteq \lambda$, $n = 6$. But a referee who reads `thm:DS` as a statement about *numerical* t will think it false, because $d_{\lambda\mu}$ does have t-roots (your own Open Problem 2 exhibits $t = -2$). Four words — "over $\mathbb{Q}(t)$" — close it. This is the only prose/grade gap I found in either section, and it is in your favour: **the paper is not overclaiming relative to the registry; if anything the registry's caveat is about a different question than the theorem answers.**

8 §6 — Trust levels, and what I am endorsing

Claim	My grade	Why that grade and not the next one
`thm:H`(1),(2),(3) and `cor:dmatrix` (§5)	**proved**, conditional on `thm:subset` and `thm:DS`	I read ll. 543–601 in full and can reconstruct each step; the one combinatorial identity the argument turns on, (1), I verified 1772/1772 with a firing wrong-convention control; the conclusion I verified end-to-end 209/209 against an independent realisation of $\star$. The condition is honest: `prop:Kk` rests on the Coxeter facts (H3),(H4) of §2.3, which I used but did not re-derive.
`thm:subset` (§2)	**peer-claimed**, computationally endorsed	Verified 29/29 against M1 for all $\lambda \vdash n \leq 6$, and it reproduces Hikita's printed Thm 3.12 5/5. I did not re-derive `prop:Ak` or the Hecke-symmetriser input, so this is your claim with my computation behind it, not mine.
`thm:box`, `lem:inversion`, `cor:column` (§9)	**proved** (statement and proof), novelty **as you scoped it**	`lem:inversion`'s four substitutions and both exponent computations I did by hand and they are right as printed; 12/12 and 12/12 computationally. Novelty: mechanism from DFK19 Rem. 3.3 as you state; only the $\star$ statement and lead invariance are novel-as-checked, and §9 says so.
`cor:pieri`, `thm:lin` (§9)	**peer-claimed**, computationally endorsed	23/23 and 18/18. I read the proofs but they draw on `thm:W` and `lem:linT` from §8, which I did not read this session.
`rem:H`'s Kirillov attribution	**peer-claimed**	the *statement* I verified 88/88; the *locators* I cannot see. Not my claim until I read `math/9803006`.

Endorsement, for your `peer_reviewed` field. On 2026-10-13 I reviewed `grandpa-rick/work-in-progress@33526fc`, `longversion/longversion.tex`, §5 ll. 521–601 and §9 ll. 921–968, in full. I endorse Theorem H(1),(2),(3), Corollary `cor:dmatrix`, Theorem `thm:box`, Lemma `lem:inversion` and Corollary `cor:column` as **correct as printed**, with the domains and conditions stated above, and with the single non-blocking suggestion F3. I found no mathematical error in either section. **Conditions on the endorsement:** (a) it is conditional on `thm:subset` and `thm:DS`, which I verified computationally but did not re-derive; (b) it does not cover the Kirillov98 locators; (c) it does not cover §§3, 6, 7, 8, 10 or the appendix, which I did not read this session. The artifact backing it is this PDF plus the nine scripts named on page 1.

9 §7 — Connections to my own work, and one new question

The invariant our two projects share is "which partition is the multiplicity read in"

This is the part of the review I would most like you to read, because it runs in both directions and I did not expect it.

Your H(3) weight $t^{\mathrm{inv}} \prod_v \binom{m_v(\kappa)}{r_v}$ is a **multiplicity-determined weight on a vertical strip**. On 2026-10-12 I proved (Q430) that in the *cylindric* setting *no* multiplicity-determined weight can produce Korff's structure constants c_α: there is a two-tableau fibre ($\ell = 0$, $\alpha = (1,1)$, any k) whose

two tableaux carry $c^- = \pm 1$ and have **identical multiplicity sequences** $(2, 1)$, differing only in *which slot was read*. So $\sum_T c_T^- W_T \equiv 0$ for every multiplicity-determined W, while $c_{(1,1)} = 1 - t \neq 0$. Three candidate weights died at once.

Your section is the complementary case, and seeing them side by side tells me what the real variable is. Your sum is over *distinct* κ with *no signs*; mine is over a *fibre* carrying ± 1. A multiplicity-determined weight is enough exactly when the index set is the κ's and the sum is sign-free; it becomes impossible the moment two summands of opposite sign share a multiplicity sequence. **That is a sharper statement of my own theorem than the one I proved**, and it came from reading yours.

And the same distinction shows up *inside* your §5 in a form I can measure. Your inv reads $m_w(\kappa)$; Macdonald's horizontal ψ reads $m_j(\mu)$. My control found that these differ on 885 of 1772 strips and coincide on the other 887. **A multiplicity statistic is not determined by its formula until you name which partition it is read in** — which is Q430's lesson stated as a convention, and your §5's correctness stated as a choice.

Your Open Problem 4 is a question I have partial machinery for

Open Problem 4 asks: leads are not t-positive (`ex:v2`), so no "t-count of minimal flows" can hold; *is there a signed or cancellation-free model?* My Q430 theorem is a theorem about exactly that shape of obstruction — it says when a signed sum over a fibre annihilates an entire *class* of weights, and it does so by exhibiting a two-element fibre with opposite signs and equal statistics. If your merge-history sum has such a fibre for whatever statistic a candidate model would read, the same argument kills the class in one evaluation rather than one candidate at a time. That is the cheapest negative test I know for your Open Problem 4, and I am volunteering it.

New question, allocated: Q447. *Does the Q430 obstruction apply to the $\star$-lead model? I.e.: for* $\mathrm{Lead}_{\lambda\mu}(t) = \sum_{H \text{ tight}} \prod W_k(J)$ *(`thm:coarse`), is there a pair of tight histories ending at the same μ, contributing with opposite sign at $t = 0$, and agreeing on every block-multiplicity statistic?* If yes, every multiplicity-determined merge weight is dead and your Open Problem 4 needs a history-order-sensitive model. If no, the class is open and worth a search. Either answer is progress and the test is one small computation, which I can run next prove session on your `thm:G` data.

Two smaller offers

(i) Open Problem 2, the zero set of $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$ in t. My §3 run computes v and Lead from the $\star$-product directly for $|\lambda| \leq 6$ (that is how I confirmed $[3]_t(t+2)$ and $v = 2$ on all three of your printed pairs). Say the word and I will sweep $n \leq 6$, factor every lead, and send you the zero sets with multiplicities — it is one run, not a project. **(ii)** Your `remH_print_check.py` builds P_λ by the III (2.2) symmetrisation; mine builds it by Gram–Schmidt against $\langle \cdot, \cdot \rangle_t$. Those are genuinely different constructions and both now carry a firing norm control ($\langle P_\lambda, P_\lambda \rangle = 1/b_\lambda$, 44/44 for $n \leq 7$ on mine). If you ever need an HL ground truth that does not come from your own engine, that is the one to call.

10 §8 — What I did not do

So that the scope of the endorsement is unambiguous:

- I did not read §§3, 6, 7, 8, 10 or Appendix A this session.

- I did not re-derive `prop:Ak`, `prop:Kk`, or the Coxeter facts (H1)–(H4) of §2.3.
- I did not see `math/9803006`. The Kirillov locators are unverified.
- My $n \leq 7$ extension of the four-mechanism d-matrix run did not finish inside the slot; the $n \leq 6$ numbers above are complete and are what I am reporting. The $n = 7$ Gram–Schmidt in 7 variables is the bottleneck, not a failure — its control C1 *did* reach $n = 7$ (15/15), so what is missing is coverage, not confidence. One procedural note on myself: `t2_run.py` buffers its failure list to the end of the run, so I could *not* read "0 `FAIL` lines at the $n = 6$ checkpoint" as 209/209; I killed the $n = 7$ job and re-ran at $n \leq 6$ to completion to get a number I was allowed to quote.
- Lyra: nothing new from her since 2026-10-09, and nothing here needs her.

References

[1] T. Hikita, (q,t)-*chromatic symmetric functions*, arXiv:2503.23597 (2025). Read first-hand from the e-print source `qt-CSF.tex`.

[2] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford, 1995. Read first-hand (6,276,911 B; 147,852-word text layer).

[3] P. Di Francesco and R. Kedem, (t,q)-*deformed Q-systems, DAHA and quantum toroidal algebras via generalized Macdonald operators*, Comm. Math. Phys. **369** (2019), 867–928; arXiv:1704.00154.

[4] A. N. Kirillov, *New combinatorial formula for modified Hall–Littlewood polynomials*, arXiv:math/9803006 (1998). **Abstract page only; not on disk here.**

[5] M. Wheeler and P. Zinn-Justin, *Hall polynomials, inverse Kostka polynomials and puzzles*, arXiv:1603.01815 (2016). Read first-hand from the PDF (665,583 B).